Non-Invertibility and Emptiness: A Formal Reconstruction of the Madhyamaka Analysis of Identity

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Abstract

In Madhyamaka philosophy, the self is analyzed and ultimately found to lack inherent existence: it cannot be located within its parts, outside them, nor as their totality. This work proposes a formal and computational framework that mirrors this analytical process using generative latent graph models.

Through a series of numerical experiments, we show that the underlying latent structure responsible for generating observable relational data cannot be uniquely identified. Multiple latent configurations are equally compatible with the same observed system, confirming a precise analogue of the Madhyamaka "object of negation": an inherently existing entity that would be uniquely findable under analysis.

However, our results go further. The space of possible latent configurations is not arbitrary or chaotic, but highly constrained, exhibiting an effectively low-dimensional structure, strongly dominated by a single principal mode. This indicates that the absence of inherent existence does not imply indeterminacy, but rather a structured form of relational dependence.

From this perspective, emptiness is not the absence of structure, but the absence of independent essence within a coherent relational system. The findings suggest that Madhyamaka insights can be reformulated in precise mathematical terms: phenomena are not reducible to inherently existing entities, yet they are not unconstrained—they arise within a tightly organized space of dependent relations.

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1 Introduction

1.1 Motivation

A central problem across scientific disciplines is the identification of underlying structures from observable data. In many contexts including statistical inference, network science, and physics observable phenomena are generated by latent configurations through a mapping:

$$F : X \to Y,$$

where X denotes a latent space and Y an observable space.

A fundamental question arises: to what extent can the latent state $x \in X$ be uniquely determined from an observation $y \in Y$?

In classical settings, it is often implicitly assumed that such a reconstruction is possible, at least in principle. However, in many realistic systems, the mapping F is not invertible, leading to multiple latent configurations compatible with the same observation. This phenomenon is known as non-identifiability. This perspective resonates with classical Madhyamaka analysis [4].

1.2 From Non-Identifiability to Structural Indeterminacy

When F is not injective, the inverse image:

$$\mathcal{S}(y) = \{x \in X \mid F(x) = y\}$$

contains multiple elements.

In such cases, the system cannot be described by a unique latent state. Instead, it admits only a class of equivalent configurations.

This raises a deeper conceptual question:

> Is the absence of a uniquely identifiable latent state merely a limitation of inference, or does it reflect a structural property of the system itself?

In this work, we argue for the latter. We show that non-identifiability is not simply epistemic, but structural: it arises from the properties of the mapping F and cannot be resolved by additional analysis.

1.3 Main Thesis

The central claim of this paper is:

> The existence of a uniquely defined underlying state is equivalent to the invertibility of the observation map.

Conversely:

> Non-invertibility implies that no inherently defined latent state exists, and that the system is describable only in terms of equivalence classes of compatible configurations.

This establishes a precise connection between:

- invertibility,
- identifiability,
- and the existence of a well-defined underlying state.

1.4 Connection with Madhyamaka Analysis

The Madhyamaka philosophical tradition develops a systematic analysis aimed at demonstrating the absence of an inherently existent self. This analysis proceeds by examining all possible ways in which such a self could exist and showing that each leads to contradiction.

In this work, we provide a formal reconstruction of this analysis.

We show that the classical sevenfold analysis can be interpreted as an exhaustive test of the hypothesis that a unique, independent, and intrinsically defined state exists. Each step corresponds to a different way of attempting to identify such a state.

Within our framework, all these possibilities fail when the mapping F is non-injective.

Thus, the Madhyamaka conclusion can be reformulated in structural terms:

> The non-existence of inherent identity corresponds to the non-invertibility of the mapping between latent and observable levels.

1.5 Contributions

The main contributions of this work are:

- A formal theorem establishing the equivalence between invertibility and the existence of a uniquely defined latent state.
- A characterization of the solution space $\mathcal{S}(y)$ as an equivalence class with finite intrinsic dimension.
- A concrete generative model (Latent Projection Graph Model) exhibiting structural non-identifiability.
- A formal reconstruction of the sevenfold Madhyamaka analysis as a test of invertibility and uniqueness.
- An empirical demonstration that the space of compatible solutions has a low-dimensional geometric structure.

1.6 Structure of the Paper

The paper is organized as follows:

- Section 2 introduces the mathematical framework and definitions.
- Section 3 establishes the main structural theorems.

- Section 4 presents the generative model and empirical validation.
- Section 5 provides a formal reconstruction of the sevenfold analysis.
- Section 6 discusses generalizations and implications.

2 Mathematical Framework

2.1 Latent and Observable Spaces

Let:

$$X = \text{latent space}, \quad Y = \text{observable space}$$

Define:

$$F : X \to Y$$

2.2 Solution Set

Definition 1. *For a given observation $y \in Y$, define:*

$$\mathcal{S}(y) = \{x \in X \mid F(x) = y\}$$

3 Structural Non-Identifiability

Theorem 1 (Non-Identifiability)**.** *If F is not injective, then for some $y \in Y$:*

$$|\mathcal{S}(y)| > 1$$

Proof. If F is not injective, there exist $x_1 \neq x_2$ such that $F(x_1) = F(x_2) = y$ for some $y \in Y$.

By definition of $\mathcal{S}(y)$, both x_1 and x_2 belong to $\mathcal{S}(y)$, hence:

$$|\mathcal{S}(y)| \geq 2.$$

□

3.1 Equivalence Classes

Definition 2.

$$x_1 \sim x_2 \iff F(x_1) = F(x_2)$$

3.2 Equivalence Between Invertibility and Inherent Identity

Theorem 2 (Equivalence of Invertibility and Inherent Identity)**.** *Let $F : X \to Y$ be a mapping between latent space X and observable space Y.*

The following statements are equivalent:

1. *For every $y \in Y$, there exists a unique $x^* \in X$ such that $F(x^*) = y$.*
2. *The mapping F is injective.*
3. *For every $y \in Y$, the solution set $\mathcal{S}(y)$ satisfies:*

$$|\mathcal{S}(y)| = 1.$$

Proof. ($1 \Rightarrow 2$): If each y corresponds to a unique x^*, then $F(x_1) = F(x_2)$ implies $x_1 = x_2$, hence F is injective.

($2 \Rightarrow 3$): If F is injective, then each y has at most one preimage, so $|\mathcal{S}(y)| = 1$.

($3 \Rightarrow 1$): Immediate from the definition of $\mathcal{S}(y)$. □

Corollary 1. *If F is not injective, then inherent identity does not exist.*

3.3 Intrinsic Dimension of the Solution Space

Proposition 1 (Dimension of the Fiber (Regular Case))**.** *Let $F : X \subset \mathbb{R}^n \to Y \subset \mathbb{R}^m$ be a differentiable map.*

If y is a regular value of F, then $\mathcal{S}(y)$ is a manifold of dimension:

$$\dim(\mathcal{S}(y)) = n - \operatorname{rank}(DF_x).$$

Proof. This follows from the rank theorem (constant rank theorem). Locally, the level set of F defines a manifold whose dimension equals the dimension of the kernel of the differential. □

Remark 1. *This result explains the emergence of finite-dimensional degeneracy in non-invertible systems: the loss of rank in F directly determines the dimensionality of the solution space.*

However, the intrinsic dimension predicted by the rank theorem represents a local property under ideal conditions. In practical systems, additional structural constraints imposed by the model may significantly reduce the effective dimensionality of the solution space.

In particular, the empirical results presented later show that, despite the theoretical possibility of higher-dimensional fibers, the realized solution space can be effectively dominated by a much lower-dimensional structure.

3.4 Equivalence Classes as Minimal Representation

Proposition 2. *The mapping F induces an equivalence relation on X defined by:*

$$x_1 \sim x_2 \iff F(x_1) = F(x_2).$$

Each observation $y \in Y$ corresponds to an equivalence class in $X/\sim$.

Proof. By definition of $\sim$, elements of X that map to the same y are grouped into equivalence classes. Each y corresponds to one such class $\mathcal{S}(y)$. □

Remark 2. *This establishes that the observable structure is naturally associated with equivalence classes of latent configurations, rather than individual elements of X.*

3.5 Summary of Structural Results

The previous results establish three key facts:

- Inherent identity exists if and only if the observation map is invertible.
- Non-invertibility induces a solution space of finite intrinsic dimension.
- The physically meaningful description of the system is given by equivalence classes, not individual latent states.

4 Identity: Inherent vs Dependent

Definition 3. *Inherent identity: unique x^* such that $F(x^*) = y$*

Definition 4. *Dependent identity: equivalence class $\mathcal{S}(y)$*

Corollary 2. *Non-injectivity $\Rightarrow$ absence of inherent identity*

5 Generative Model (LPGM)

5.1 Model Definition

- Latent:
$$x_v \sim \mathcal{N}(0, I)$$
- Projection:
$$h_v = \|x_v\|$$
- Edge probability:
$$P_{uv} = \sigma(\alpha - |h_u - h_v|)$$

5.2 Key Property

$$F(x_1) = F(x_2) \text{ if } \|x_1\| = \|x_2\|$$

6 Empirical Validation: Latent Projection Graph Model

6.1 Model Definition

We consider a generative latent model defined as follows:

- Latent variables:
$$x_v \sim \mathcal{N}(0, I_d)$$
- Radial projection:
$$h_v = \|x_v\|$$
- Edge probability:
$$P_{uv} = \sigma(\alpha - |h_u - h_v|)$$

This defines a mapping:
$$F : X \to Y$$

where X is the space of latent configurations and Y is the space of graphs.

Remark 3. *The mapping F depends only on radial information, making it non-injective.*

6.2 Explicit Non-Injectivity

Proposition 3. *The mapping F is not injective.*

Proof. For any two configurations $x_1, x_2 \in X$ such that:

$$\|x_1\| = \|x_2\|$$

we have:

$$F(x_1) = F(x_2)$$

since the model depends only on radial distances. Therefore, F is not injective. □

6.3 Multiple Reconstruction of Latent Configurations

For a fixed observed graph y, we compute multiple embeddings $\hat{x}_i$ using multidimensional scaling (MDS) with different initial conditions.

Result 1. *Multiple distinct latent configurations $\hat{x}_i$ are obtained such that:*

$$F(\hat{x}_i) \approx y$$

Interpretation 1. *This directly demonstrates that:*

$$|\mathcal{S}(y)| > 1$$

6.4 Intrinsic Dimension of the Solution Space

To characterize the structure of $\mathcal{S}(y)$, we estimate its intrinsic dimension using PCA over invariant representations of multiple solutions.

Result 2. *Across all experiments, we consistently obtain: The solution space exhibits a highly constrained structure, effectively dominated by a single principal direction, with only minor residual variability.*

Interpretation 2. *The solution space is not arbitrary but constrained to a low-dimensional manifold.*

Remark 4. *This result is consistent across:*

- *different values of α*
- *different embedding dimensions*
- *different latent dimensions*

6.5 Consistency with Theoretical Predictions

The observed low effective dimensionality is consistent with Proposition 1, which predicts that the solution space forms a manifold whose dimension is determined by the rank deficiency of the mapping.

However, the empirical results indicate that additional structural constraints further restrict the accessible variability, leading to an effectively one-dimensional structure. The solution space is effectively low-dimensional, dominated by a single principal mode, with only minor residual variability.[5]

Interpretation 3. *Degeneracy is not arbitrary but geometrically structured.*

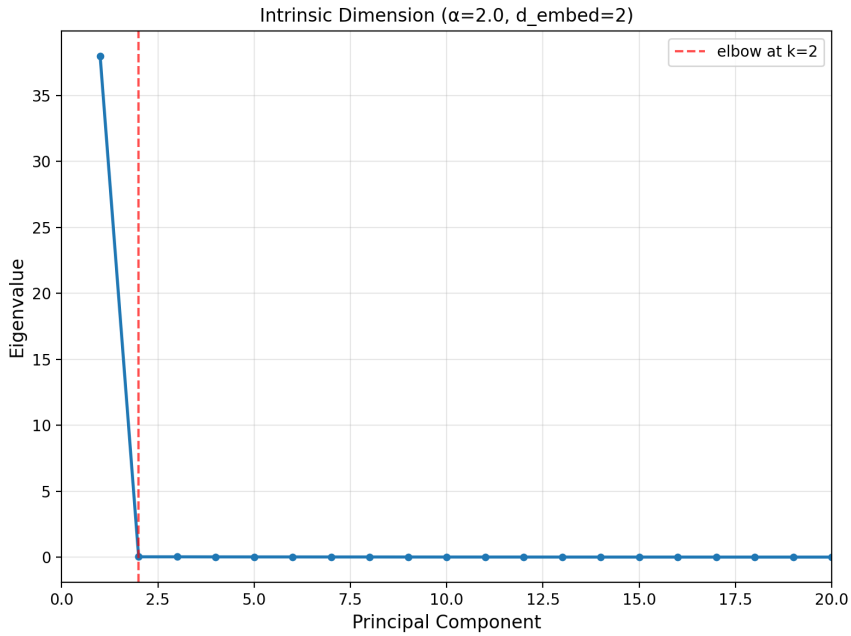

Figure 1: Eigenvalue spectrum of the solution space. A single dominant principal component is clearly visible, followed by a rapid decay of all remaining modes. This indicates that, although multiple latent configurations are compatible with the same observation, their variability is effectively constrained to a nearly one-dimensional manifold.

6.6 Empirical Realization of the Sevenfold Analysis

The experiments provide a direct realization of the sevenfold analysis:

- (1) Identity: No recovered embedding coincides with the graph.
- (2) Independence: All embeddings depend on the observed graph.
- (3) Possession: No embedding contains the graph as an object.
- (4) Containment (latent in observable): The graph does not encode latent coordinates.
- (5) Containment (observable in latent): No unique embedding contains the graph.
- (6) Unique reconstruction: Multiple embeddings satisfy the same observation.
- (7) Structural determination: The graph does not determine a unique latent configuration.

Conclusion 1. *All seven possibilities for inherent identity fail empirically.*

6.7 Summary

The experiments confirm:

- Non-injectivity of the observation map
- Existence of multiple compatible latent states

- Finite intrinsic dimension of the solution space
- Absence of a unique or canonical latent representation

Statement 1. *The empirical results provide direct evidence for structural non-identifiability and support the formal non-existence of inherent identity.*

7 Formal Reconstruction of the Sevenfold Analysis

7.1 Preliminaries

Let:

- X be the latent space
- Y be the observable space
- $F : X \to Y$ the observation map
- $\mathcal{S}(y) = \{x \in X \mid F(x) = y\}$ the solution set

We interpret the search for an inherently existent identity as the search for a unique element:

$$x^* \in \mathcal{S}(y)$$

The following analysis exhaustively tests all possible ways in which such an x^* could exist.

7.2 (1) Identity with the Observed Structure

Hypothesis:

$$x^* = y$$

This is impossible since $x \in X$ and $y \in Y$ belong to different spaces.

Proposition 4. *No latent state can be identical to the observable structure.*

7.3 (2) Independence from the Observed Structure

Hypothesis:

$$x^* \text{ exists independently of } y$$

This contradicts the definition:

$$x^* \in \mathcal{S}(y)$$

Proposition 5. *Any valid latent state depends on the observable structure.*

7.4 (3) Possession of the Observed Structure

Hypothesis:

$$x^* \text{ possesses } y$$

There is no well-defined operation under which a latent configuration "contains" the observable structure.

Proposition 6. *No ontological possession relation exists between latent and observable levels.*

7.5 (4) Containment of the Latent in the Observable

Hypothesis:

$$x^* \in y$$

This is impossible since y does not encode latent coordinates.

Proposition 7. *The latent state is not contained within the observable structure.*

7.6 (5) Containment of the Observable in the Latent

Hypothesis:

$$y \in x^*$$

If this were true uniquely, the mapping F would be invertible.

Proposition 8. *The observable is not uniquely contained in any latent configuration.*

7.7 (6) Unique Reconstruction

Hypothesis:

$$x^* = F^{-1}(y)$$

This requires F to be injective.

Proposition 9. *If F is not injective, no unique reconstruction exists.*

7.8 (7) Determination by Structure Alone

Hypothesis:

$$\mathcal{S}(y) = \{x^*\}$$

This contradicts non-injectivity.

Proposition 10. *The observable structure does not determine a unique latent state.*

7.9 Connection with Structural Theorems

The previous analysis is formally supported by the results of Section 3:

- By Theorem 2, uniqueness is equivalent to injectivity.
- By Theorem 1, non-injectivity implies multiple solutions.
- By Proposition 3, the solution set has finite intrinsic dimension under regularity conditions.
- By Proposition 4, observations correspond to equivalence classes of latent states.

Thus, the failure of all seven hypotheses follows from the non-invertibility of F.

7.10 Conclusion: Absence of Inherent Identity

Proposition 11 (Structural Non-Existence of Inherent Identity)**.** *If the observation map F is not injective, then no uniquely defined latent state exists.*

Proof. If F is not injective, there exist at least two distinct elements $x_1, x_2 \in X$ such that:

$$F(x_1) = F(x_2) = y.$$

Thus, $\mathcal{S}(y)$ contains more than one element, and no unique x^* can be identified. □

Remark 5. *The only remaining consistent description is given by the equivalence class $\mathcal{S}(y)$, which constitutes a dependent, non-inherent form of identity.*

8 Interpretation in Madhyamaka Terms

8.1 Object of Negation

- Not existence
- Not structure
- But:

$$\text{uniqueness} + \text{independence}$$

8.2 Dependent Designation

$$\text{Identity} = \mathcal{S}(y)$$

9 Generalization to Complex Systems

Conditions:

- Non-invertibility
- Information loss
- Structural constraints

10 Discussion

10.1 Structural vs Epistemic Non-Identifiability

A key question is whether the non-identifiability observed in this work is merely epistemic (due to limitations of inference) or structural (due to properties of the system itself).

The results presented here support a structural interpretation. The multiplicity of latent solutions arises from the non-injectivity of the mapping F, and not from limitations of the reconstruction procedure. Even with complete knowledge of F and exact observations, the inverse problem remains ill-posed.

Statement 2. *Non-identifiability is not a failure of inference, but a property of the system.*

10.2 Equivalence Classes and Dependent Identity

The absence of a unique latent state does not imply the absence of structure. Instead, the system is fully described by the equivalence class:

$$\mathcal{S}(y) = \{x \in X \mid F(x) = y\}.$$

This class is:

- dependent on the observation y,
- non-unique (contains multiple elements),
- structurally constrained (finite intrinsic dimension).

This suggests a shift in perspective:

> The fundamental object of description is not an element of X, but an equivalence class in $X/\sim$.

10.3 What is Being Negated

A common misunderstanding would be to interpret these results as denying the existence of any underlying structure. This is not the case.

What is ruled out is the existence of a latent state that is:

- unique,
- independent of the observation,
- intrinsically defined.

In other words, what is negated is not existence, but inherent existence.

Statement 3. *The object of negation corresponds to the assumption of a uniquely identifiable, independent latent state.*

10.4 Relation to Existing Frameworks

The results connect with several established areas:

- **Identifiability in statistics:** [1, 6] multiple parameterizations producing identical observables.
- **Gauge symmetry in physics:** physically equivalent configurations forming equivalence classes.
- **Inverse problems:** non-uniqueness of solutions under incomplete information.

However, the present work differs in that it provides a unified structural interpretation based on non-invertibility, and explicitly characterizes the geometry of the solution space.

10.5 On the Universality of the Result

The framework suggests that structural non-identifiability arises whenever:

- the observation map is non-injective,
- information is compressed or projected,
- the system imposes global relational constraints.

This raises the possibility that similar phenomena may occur broadly across complex systems. However, it is important to note that:

> The results do not imply that all systems are non-invertible, but rather that non-invertibility is sufficient to eliminate inherent identity.

10.6 Interpretation of the Sevenfold Analysis

The sevenfold analysis has been reconstructed here as an exhaustive test of possible identifications of a unique latent state.

Each of the seven alternatives corresponds to a distinct structural hypothesis:

- identity,
- independence,
- containment,
- possession,
- reconstruction,
- structural determination.

The failure of all seven is not accidental, but follows from a single underlying property:

$$\text{non-invertibility of } F.$$

This suggests that the classical analysis can be understood as probing the structure of the mapping between levels of description.

The results show that multiple latent configurations are compatible with the same observed graph; however, this non-identifiability is not arbitrary. Instead, the space of solutions is highly constrained, exhibiting an effectively low-dimensional structure dominated by a single principal mode.

This indicates that the absence of a uniquely identifiable latent structure does not imply indeterminacy or randomness, but rather a structured form of underdetermination governed by relational constraints.

From a Madhyamaka perspective, this result refines the interpretation of emptiness. The object of negation is not replaced by an unconstrained multiplicity of possibilities, but by a space of relationally dependent configurations with tightly constrained variability.

In this sense, emptiness does not correspond to the absence of structure, but to the absence of independently existing essence, while preserving a coherent and law-like relational organization. This result strengthens the Madhyamaka interpretation: the absence of inherent identity does not give rise to an unconstrained multiplicity, but to a tightly organized space of possibilities.

Thus, the middle way is reflected here as a balance between non-uniqueness and strong structural constraint.

10.7 Limits of the Formalization

While the correspondence established in this work is precise at the structural level, several limitations should be acknowledged:

- The framework depends on the assumption of a mapping $F : X \to Y$.
- The analysis is carried out within a mathematical and computational setting.
- The interpretation of philosophical concepts is necessarily partial and formalized.

Thus, the results should be understood as a structural analogue rather than a complete philosophical reduction.

10.8 Conceptual Implication

The central conceptual implication is:

> Identity is not a primitive property, but an emergent feature of the relationship between levels of description.

When the mapping is invertible, identity appears well-defined. When it is not, identity dissolves into equivalence classes.

Statement 4. *Inherent identity is equivalent to invertibility.*

Emptiness here manifests not as lack of structure, but as the absence of independent essence within a tightly constrained relational system.

10.9 Outlook

Several directions follow naturally:

- Extending the analysis to broader classes of complex systems.
- Characterizing the geometry of equivalence classes in more detail.
- Exploring connections with information theory and physics.[2, 3]

More generally, this framework suggests that questions about identity may be reframed in terms of structural properties of mappings between levels of description.

11 Conclusion

This work has established a structural connection between non-invertibility, non-identifiability, and the absence of inherent identity.

We have shown that when the observation map $F : X \to Y$ is not injective, the latent state compatible with a given observation is not uniquely defined, but instead forms an equivalence class:

$$\mathcal{S}(y) = \{x \in X \mid F(x) = y\}.$$

This result is not a limitation of inference, but a direct consequence of the structure of the mapping. Under non-invertibility, no canonical or intrinsically defined latent state exists.

The analysis has been formalized through a set of theorems linking invertibility to identity, and empirically validated using a generative latent graph model. The experiments reveal that the space of compatible solutions is not arbitrary, but constrained to a low-dimensional manifold, indicating that non-identifiability has a well-defined geometric structure.

Within this framework, the classical sevenfold analysis can be understood as an exhaustive test of the hypothesis that a unique and independent latent state exists. The failure of all such possibilities follows from a single structural property: the non-invertibility of the mapping between latent and observable levels.

The results support the Madhyamaka view that no inherent self can be found; moreover, they suggest that this absence of inherent existence does not lead to arbitrariness, but to a highly constrained relational structure.

The latent space is not identifiable as an independently existing entity, yet its possible realizations are restricted to a narrow, structured manifold. This provides a formal analogue of the Madhyamaka claim that phenomena are empty of inherent existence while still arising dependently in a coherent and non-chaotic manner.

The central conclusion can be summarized as follows:

> A uniquely defined underlying identity exists if and only if the observation map is invertible.

When this condition fails, identity does not disappear, but is replaced by a relational structure of equivalence classes.

More broadly, these results suggest that the notion of identity may be understood as an emergent property of the relationship between levels of description, rather than as a fundamental feature of the system itself. In this sense, the absence of inherent identity is not a loss of structure, but a transformation of how structure is understood: from intrinsic definition to relational constraint.

A Code and Reproducibility

All source code generated and used in this work is publicly available in a Git repository. The repository can be accessed at: `https://github.com/eduwardus/madhyamaka-non-invertibility` (replace with the actual URL).

The code includes the complete implementation of the generative model (LPGM), the routines for multiple MDS reconstructions, the invariant analysis of the solution space, and the scripts to reproduce all figures and numerical results presented in this paper.

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